

Exercise 1.i

Finding the equilibrium analitically:

```
syms c(t) V Q k F;  
ode(t) = diff(c(t),t) == (F/V)-(Q*c/V)-(k*c^2);  
F=200;V=10;Q=1;k=0.1; % to subs in symbolic function  
ode(t)=subs(ode(t))
```

ode(t) =

$$\frac{\partial}{\partial t} c(t) = -\frac{c(t)^2}{10} - \frac{c(t)}{10} + 20$$

If we let $c(t) = x$, then we can solve for $\frac{dc}{dt} = -\frac{c^2}{10} - \frac{c}{10} + 20 = -\frac{x^2}{10} - \frac{x}{10} + 20 = 0$:

```
syms g(x);  
g(x) = -x^2/10-x/10+20
```

g(x) =

$$-\frac{x^2}{10} - \frac{x}{10} + 20$$

```
solve(g,x);  
conc_equilib = (3*sqrt(89)/2)-0.5
```

conc_equilib = 13.6510

```
cond = c(0) == 0; % initial concentration  
conc(t)=dsolve(ode(t),cond) % analytical solution
```

conc(t) =

$$\frac{3 \sqrt{89} \tanh\left(\frac{\sqrt{89} \left(267 t + 20 \sqrt{89} \operatorname{atanh}\left(\frac{\sqrt{89}}{267}\right)\right)}{1780}\right)}{2} - \frac{1}{2}$$

```
f=@func;  
[a,b]=ode45(f,[0:0.5:15],0);  
numerical = runge_kutta(f,[0:0.1:15],0);
```

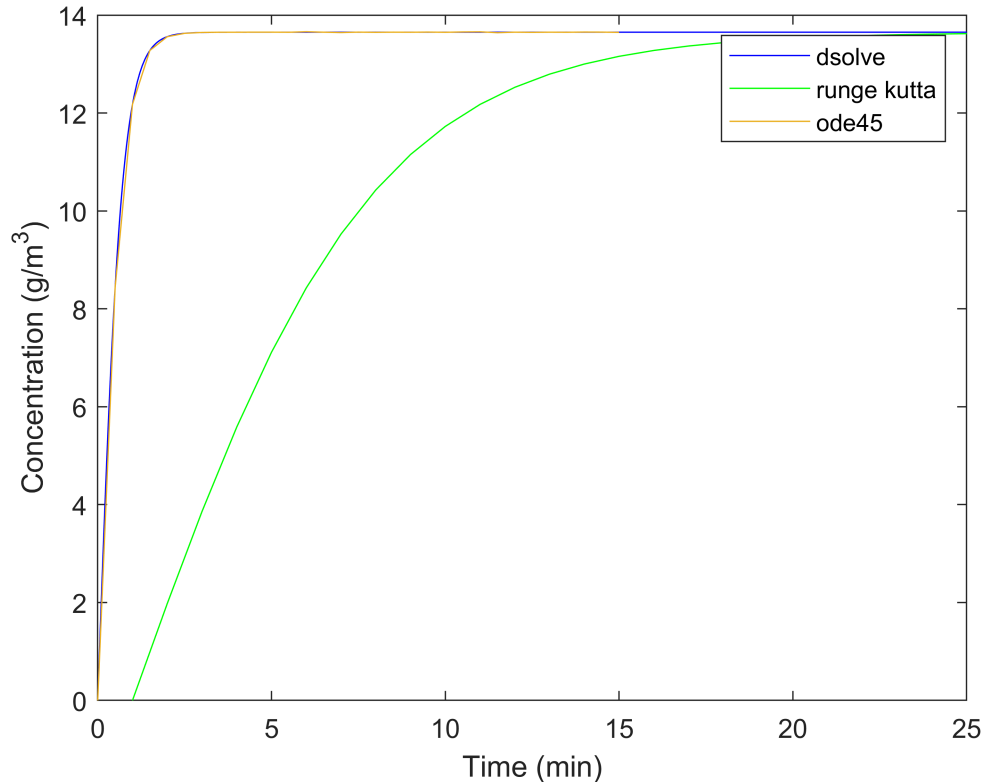
```
fplot(conc,[0 25], 'b')  
xlim([0 25]);  
ylim([0 14]);  
hold on
```

```
plot(numerical, 'g')  
xlim([0 25]);  
ylim([0 14]);  
hold on  
plot(a,b)  
xlim([0 25]);  
ylim([0 14]);
```

```

legend("dsolve","runge kutta","ode45");
xlabel('Time (min)');
ylabel('Concentration (g/m^3)');

```



The steady state concentration is $13.6510 \frac{g}{m^3}$. By the plot above, note that the analytical solution grows and approach the steady state concentration more rapidly than the Runge Kutta algorithm. However, the Runge Kutta algorithm does approach the steady state later in time.

```

function [y] = runge_kutta(f,x,y0)
n=length(x);
m=length(y0);
y=zeros(m,n);
y(:,1)=y0; % the first column is equal to the initial value vector
for i=1:n-1 % for i from 1 to length of IV - 1
    h = x(i+1)-x(i); % cal
    xi=x(i);
    yi=y(:,i);
    k1 = f(xi,yi);
    k2 = f(xi+0.5*h,yi+0.5*k1*h);
    k3 = f(xi+0.5*h,yi+0.5*k2*h);
    k4 = f(xi+h,yi+k3*h);
    y(:,i+1) = y(:,i) + h*(k1+2*k2+2*k3+k4)/6; % this is a matrix
end
end

function [dcdt] = func(t,c)

```

```
V=10;F=200;Q=1;k=0.1;  
dcdt=zeros(1,1);  
dcdt(1) = (F/V)-(Q*c./V)-k.*c.^2;  
end
```